

Rasika Edirisinghe

Marie Curie Early Stage Researcher · University College Cork, Ireland

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Summary

AI researcher specialising in Natural Language Processing, Machine Learning, and Transformers, with a strong foundation in applied data visualization. Currently pursuing a fully funded PhD in Computer Science at University College Cork under the CASCADE Horizon Europe MSCA programme, developing LLM-powered text reuse detection systems for diachronic corpora and low-resource textual domains. Over four years of cross-functional industry experience building cloud-based BI tools and automation pipelines. Committed to delivering explainable, human-centered AI solutions for language and data-intensive environments.

Education

PhD in Computer Science (Reading)

University College Cork, Ireland

Sep 2024 – Sep 2027 (expected)

CASCADE Horizon Europe MSCA

Thesis: Language Modelling and Visualization for Parallel Text Analysis

- Designing advanced text reuse detection systems for diachronic corpora and low-resource textual domains using LLMs and interactive visual analytics.
- Addressing orthographic variation, semantic drift, and context-dependent alignment across time and textual use cases.

M.Sc. Business Analytics — Distinction

Robert Gordon University, Aberdeen, UK

Sep 2021 – Sep 2023

Thesis: Twitter Sentiment Reason Mining Framework to Identify Major Problems in US Healthcare Industry (Grade: A)

Key modules: Data Management (A) · Web Mining (A) · BI Tools & Applications (A) · Intro to Big Data & Data Science (A)

B.Sc. (Hons) Business Information Systems

Cardiff Metropolitan University, Cardiff, UK

Nov 2016 – Oct 2018

Thesis: Computer-Aided Diagnosis System for Lung Cancer Detection (Java-based CADe system)

BTEC Higher National Diploma, Software Development

Pearson Edexcel, UK

May 2014 – May 2016

OOP · Data Structures & Algorithms · Networking Technologies · Web Applications · Database Development

Academic Experience

Teaching Assistant & Co-Lecturer — Data Visualization for Machine Learning

University College Cork

2025/26

CS3205 / CS6426

- Revamped lecture and lab materials for the 2025/26 cohort.
- Delivered lectures and hands-on Python lab sessions.
- Created all lecture slide decks for 2026 delivery.
- Assessed and graded student assignments and examinations.

Skills: Data Visualization, Machine Learning, Python, University Teaching, Assessment

Visiting Lecturer

University of Ruhuna, Dept. of Electrical & Information Engineering

Aug 2023 – Sep 2024

Galle, Sri Lanka

- **Introduction to Data Visualization with Power BI** — DAX, dashboards, Gestalt principles, color theory.
- **Excel for Data Science** — PivotTables, time-series, statistical analysis, Solver.
- **Data Preprocessing & Cleaning** — missing data, outlier detection, encoding.
- **Data Splitting & Cross Validation** — overfitting, metrics, confusion matrix.

Skills: Data Science Teaching, Power BI, Excel Analytics, ML Foundations

Visiting Lecturer

Sri Lanka Telecom Training Centre, Sri Lanka Telecom PLC

Aug 2023 – Sep 2024

Welisara, Sri Lanka

- **Fundamentals of AI & Intelligent Systems** (Pearson BTEC K/618/5660) — AI types, NLP, conversational AI, ethics.
- **Big Data & Visualization** (J/618/6251) — Python, Jupyter, Power BI, DAX, data governance.
- **Business Intelligence** (M/615/1641) — BI systems, OLAP, descriptive & predictive analytics.

Skills: AI Education, Big Data, Business Intelligence, Higher Education Teaching

Industry Experience

Data Analyst

EFL Global

May 2023 – Sep 2024

Colombo, Sri Lanka (Hybrid)

- Automated global finance reporting across 39 countries & 70+ offices (Python, VBA, Advanced Excel).
- Modelled and maintained Azure Databricks backend for automated MIS, AR & AP reporting.
- Built Power BI & Tableau dashboards; implemented workflows via Power Automate & Logic Apps.
- Managed Azure Analysis Services, Data Lakes, Blob Storage, and Synapse Analytics.

Skills: Python, Azure Databricks, Power BI, Tableau, Power Automate, Azure Synapse

Implementation Engineer

Affno Asia Pacific

Nov 2022 – May 2023

Colombo, Sri Lanka

- Designed and deployed BI solutions for supply chain analytics.
- Acted as Technical Business Analyst, translating business needs into system designs.
- Provided training and support to clients in reporting and analytics.
- Worked with Amazon EC2 & MongoDB Atlas for cloud-based solutions.

Skills: Business Intelligence, Supply Chain Analytics, MongoDB, Business Analysis

Operations Analyst

SYNERGEN Health

Sep 2019 – Nov 2022

Colombo, Sri Lanka

- Managed Qlik-Sense cloud-based RCM BI platform (Doctrix) for real-time KPI & SLA monitoring.
- Led process training for multiple clients in Revenue Cycle Management.
- Implemented Python & VBA automation for financial reporting and data integrity.
- Maintained Amazon DynamoDB & Kinesis Data Streams for real-time analytics.

Skills: Healthcare BI, QlikSense, Revenue Cycle Analytics, Python, VBA

Associate Operations Analyst (Intern)

SYNERGEN Health

Mar 2019 – Sep 2019

Colombo, Sri Lanka

Client reporting, process monitoring, data sanity checks, and trade support.

Business Analyst (Intern)

Global System Solutions

Sep 2018 – Mar 2019

Colombo, Sri Lanka

Requirement gathering, use case development, reporting specification, pre-sales support.

Publications & Research Output

Rewriting Tradition: Quantifying Change in Lady Gregory's Irish Legends

Rachel McCarthy, Rasika Edirisinghe, James O'Sullivan, Cliona Ó Gallchoir, Rosane Minghim, Órla Murphy
DH2025: Alliance of Digital Humanities Organizations · Lisbon, Portugal · Jul 2025 · [Abstract Accepted](#)

Sentiment Reason Mining Framework for Analyzing Twitter Discourse on Critical Issues in US Healthcare Industry

R. Edirisinghe, P. P. G. Dinesh Asanka

International Conference on Smart Computing and Systems Engineering 2023 · Aug 2023

ieeexplore.ieee.org/document/10215010

Work in Progress

- **Text Reuse Detection with Large Language Models** — PhD research, UCC (CASCADE MSCA)
- **Interactive Visualization for NLP Interpretability** — open-source tool, PhD research, UCC
- **Parsing Hiberno-English Dictionary Entries: A Hybrid Rule-Based and LLM Approach** — under preparation; presentation accepted at *Digital Humanities Congress 2026*, University of Sheffield, Sep 2–3 2026
- **Trust & Agency in Agentic Cyber-Physical Systems** — Nimbus Research Centre, inTrusted project

Presentations & Academic Contributions

Beyond Translation: Rethinking ‘Parallel Text’ Across Disciplines

UCC Digital Humanities Colloquium · University College Cork · 1 October 2025

Language Modelling and Visualization for Parallel Text

CACSSS PhD Lightning Talks · University College Cork · 4 April 2025

Sentiment Reason Mining Framework ... (paper presentation)

International Conference on Smart Computing and Systems Engineering 2023 · June 2023

Peer Reviewer — 8th International Research Conference on Smart Computing and Systems Engineering (SCSE 2025), Feb 2025

Evaluator — System Development Project Poster Presentation, University of Kelaniya, Oct 2023

Curriculum Reviewer — Foundation to Data Science, University of Ruhuna, Aug 2023

Skills & Technologies

Languages	Python, SQL, Java, JavaScript, C#, R, VBA
NLP & LLMs	HuggingFace Transformers, spaCy, Gensim, NLTK, OpenAI API, Prodigy
ML / DL	PyTorch, TensorFlow, Keras, scikit-learn, XGBoost
Visualization	Power BI, Tableau, QlikView/Sense, Plotly, Matplotlib, Seaborn, DAX
Cloud	Azure (Databricks, Synapse, Blob, Logic Apps), AWS (EC2, DynamoDB, Kinesis), MongoDB Atlas
Databases	MySQL, MongoDB, Amazon DynamoDB
Dev Tools	Git, Jupyter, Weka, Orange, HTML/CSS, ASP.NET, Power Automate, Bash
Soft Skills	Technical communication with non-technical audiences, tutoring & mentoring, business process design, collaborative team player

Current Secondment

Research Secondment — inTrusted Project

Apr 2026 – Jul 2026

Nimbus Research Centre

Cork, Ireland

Working with Tharindu Ranathunga on the [inTrusted](#) project — exploring trust frameworks for agentic cyber-physical systems, and the gap between human intentionality, AI interpretation, and real-world actions.